

UniTS Anomaly Ratio Sweep

STPSat4-PCE2 | Generated 2026-03-12 10:53

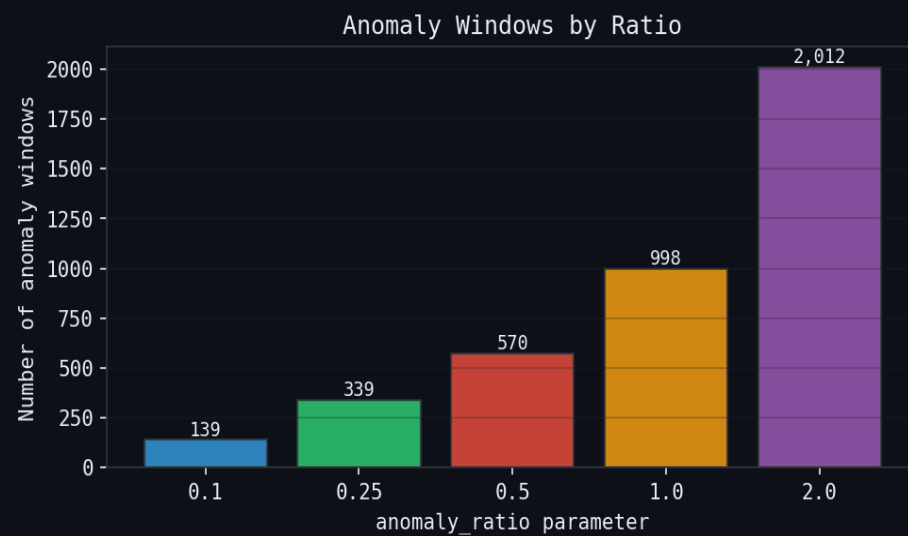
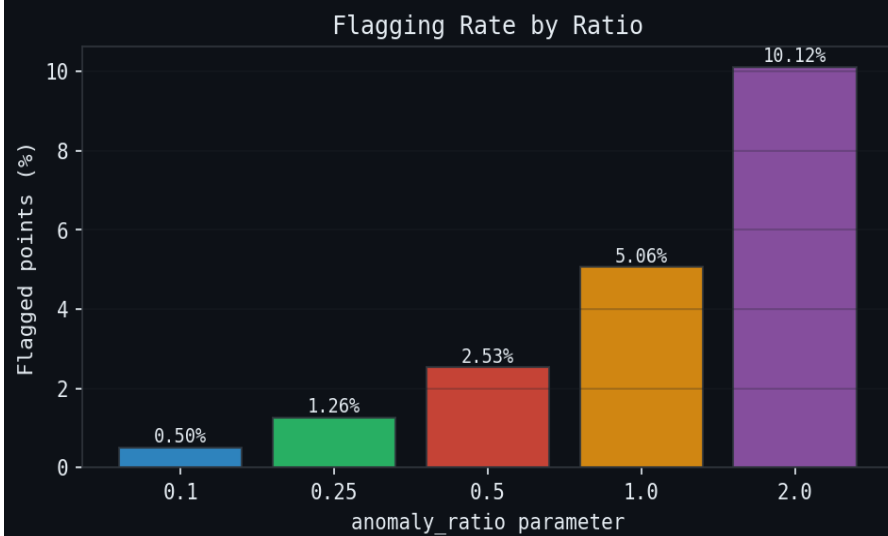
Results by Ratio

anomaly_ratio	Threshold	Flagged Points	Flag Rate	Windows
0.1	1.490112	218	0.505%	139
0.25	0.880979	546	1.264%	339
0.5	0.502089	1,093	2.530%	570
1.0	0.256044	2,186	5.060%	998
2.0	0.154120	4,372	10.120%	2,012

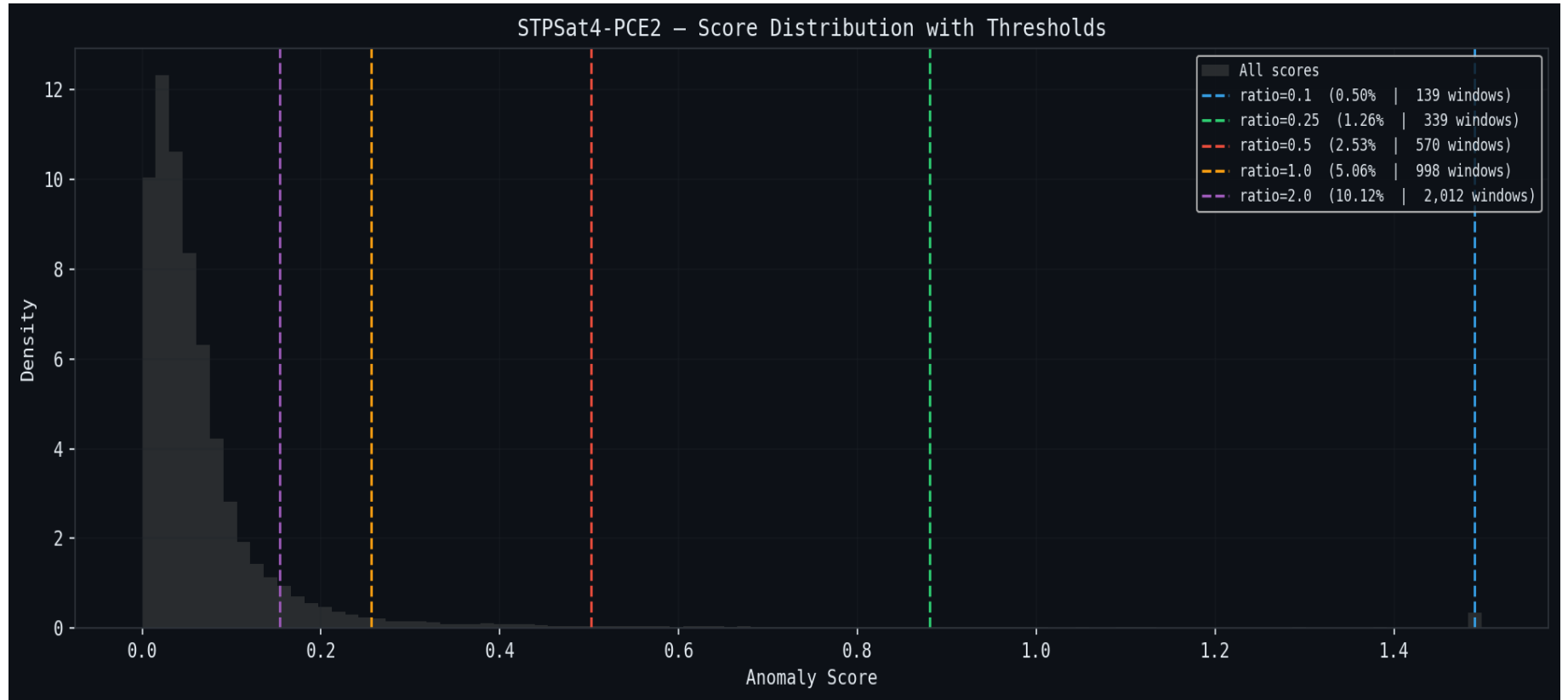
anomaly_ratio controls the detection threshold: UniTS flags the top ratio% of reconstruction error scores as anomalies. Lower values = stricter threshold = fewer flags. The threshold column shows the exact score cutoff for each ratio.

Flagging Rate & Window Count

STPSat4-PCE2 – Ratio Sweep Summary



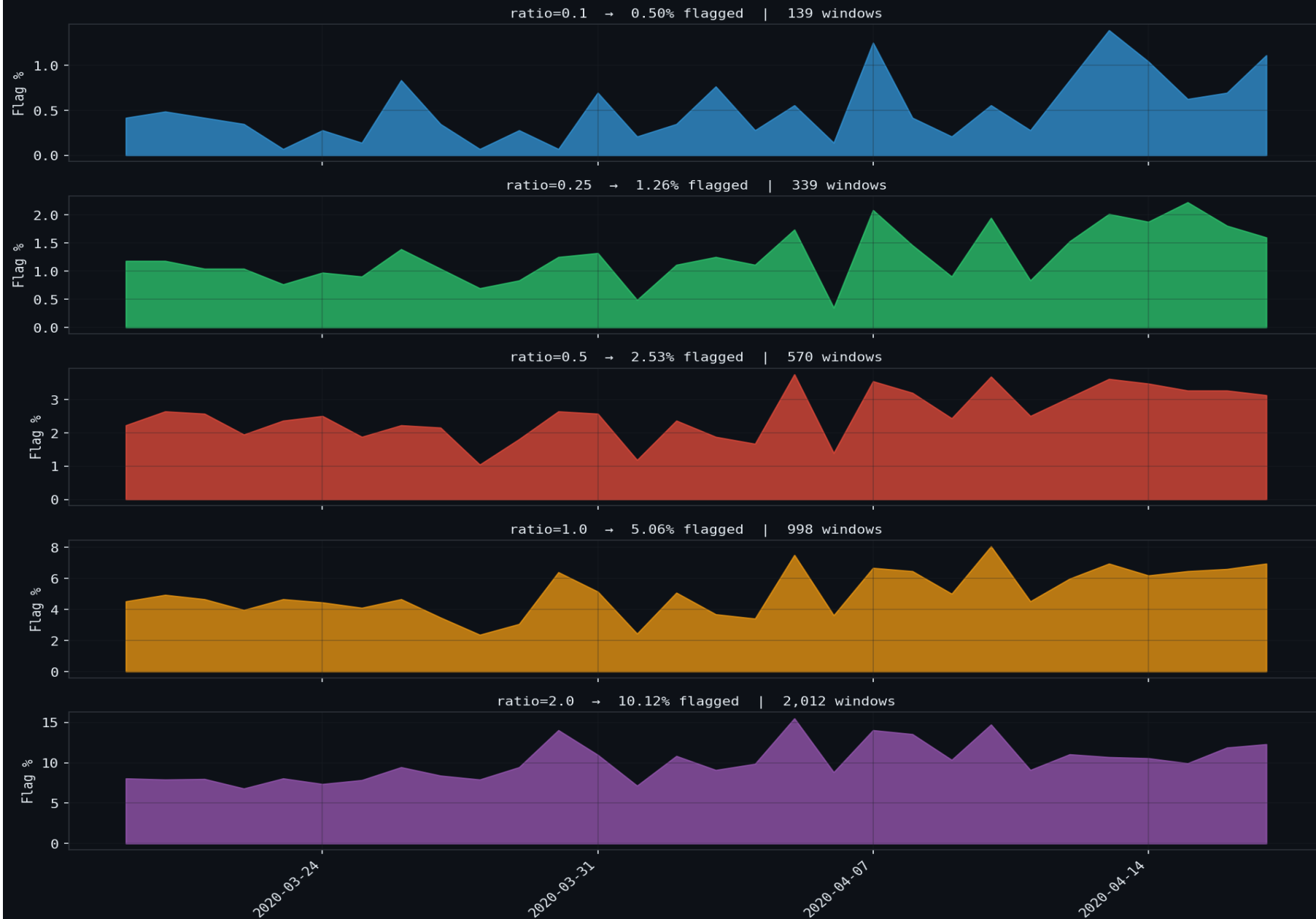
Score Distribution with Thresholds



Dashed vertical lines show where each ratio sets its threshold. Choose a ratio whose threshold sits clearly in the tail of the distribution rather than in the main body of normal scores.

Daily Flagging Rate by Ratio

STPSat4-PCE2 – Daily Flagging Rate by Ratio



A realistic anomaly rate for a nominal satellite is typically 0.1–1%. Look for a ratio that shows isolated spikes rather than sustained elevated flagging.